

Andrew Chio

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Education

- ◇ **Ph.D. in Computer Science** Jun 2026
University of California, Irvine
Thesis: *Leveraging Physics-based Models and Data-driven Methods for Cyber-Physical Human Infrastructure Resilience*
Advisor: Prof. Nalini Venkatasubramanian
- ◇ **B.S. in Computer Science, specialization in Algorithms** Jun 2019
University of California, Irvine
Magna Cum Laude – Graduated in top 4% of class

Professional Experience

- ◇ **Center for Nonlinear Studies Postdoctoral Fellow, Los Alamos, NM** Aug 2026 – Present
T-5 Group, Theoretical Division, Los Alamos National Laboratory
- ◇ **Ph.D. Researcher, Los Alamos, NM** Apr 2025 – Aug 2026
T-5 Group, Theoretical Division, Los Alamos National Laboratory
- ◇ **Graduate Student Researcher, Irvine, CA** Sep 2019 – Jun 2026
*Distributed Systems Middleware Group, Information Systems Group
Department of Computer Science, University of California, Irvine*
- ◇ **Graduate Student Internship, Los Alamos, NM** Jun 2024 – Sep 2024
T-5 Group, Theoretical Division, Los Alamos National Laboratory
- ◇ **Ph.D. Visiting Research Fellow, Los Alamos, NM** Apr 2022 – Apr 2024
T-5 Group, Theoretical Division, Los Alamos National Laboratory
- ◇ **Undergraduate Research Assistant, Irvine, CA** Sep 2018 – Jun 2019
*Distributed Systems Middleware Group, Information Systems Group
Department of Computer Science, University of California, Irvine*

Honors, Awards & Achievements

- ◇ **CNLS Fellow Appointment** Aug 2026
Center for Nonlinear Studies, Los Alamos National Laboratory
- ◇ **NSF CPS Rising Star 2025** Mar 2025
Cyber-Physical Systems Rising Stars Workshop, National Science Foundation
 - Awarded to 30 graduate students (out of 174 applicants) across the US
- ◇ **Best Poster Award, Resilience for Large-Scale Cyber-Physical Human Infrastructures** May 2025
Ph.D. Forum, ACM/IEEE 2025 CPS-IoT Week
 - Awarded to 5 graduate students participating in the PhD Forum

- ◇ **NSF-ASI Japan Fellow** May 2024
Advanced Studies Institute, National Science Foundation
 - Awarded to 12 graduate students across the US
- ◇ **ARCS Foundation Scholar** Sep 2022
Orange County Chapter, ARCS Foundation
 - Awarded to 10 graduate fellows from UC Irvine
- ◇ **UC National Lab In-Residence Graduate Fellowship** Apr 2022 – Apr 2024
Los Alamos National Laboratory, University of California Research Initiatives
 - Awarded to 7 graduate fellows across all University of California campuses
- ◇ **Mark Weiser Best Paper Award** Mar 2022
20th IEEE International Conference on Pervasive Computing and Communications (PerCom)
- ◇ **Dean's Honor Roll** Sep 2015 – Jun 2019
University of California, Irvine
- ◇ **SURF-IoT Summer Fellowship** Jun 2018 – Aug 2018
Undergraduate Research Opportunities Program, University of California, Irvine
 - Awarded to 10 undergraduates from UC Irvine
- ◇ **Most Startup Potential: MediPal** Nov 2016
Med AppJam, University of California, Irvine

Publications

Peer-Reviewed

- [14] Ryan Hildebrant, **Andrew Chio**, and Nalini Venkatasubramanian. “Synthetic Data Generation for AI-enabled Smart Cyber-Physical Infrastructures”. In: *ACM SIGMOD Workshop on Synthetic Data Generation and Management for Building AI Systems (SynthAI), Poster Presentation*. 2026.
- [13] Modeste M. Kenne, **Andrew Chio**, ZhengHui Hu, Julie Rousseau, and Nalini Venkatasubramanian. “Black-Ops: GIS-Informed Planning for Healthcare Resilience Amid Power Grid Outages”. In: *Proceedings of the 12th IEEE International Conference on Smart Computing (SMARTCOMP'26)*. 2026.
Accepted for Publication, BICORE2026.
- [12] Fernanda Ventrone, **Andrew Chio**, Ruben Vescovo, Erick Mas, Shunichi Koshimura, and Nalini Venkatasubramanian. “PARA-FLOW: An Event-Based Framework for Modeling Accessible Multi-Modal Urban Mobility”. In: *Proceedings of the 34th International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*. 2026.
Accepted for Publication, AICORE2026.
- [11] **Andrew Chio**, Russell Bent, Kaarthik Sundar, and Nalini Venkatasubramanian. “SEQUIN: A Network Science and Physics-based Approach to Identify Sequential Nk Attacks in Electric Power Grids”. In: *Proceedings of the ACM/IEEE 16th International Conference on Cyber-Physical Systems (with CPS-IoT Week 2025) (IC-CPS'25)*. 2025, pp. 1–12.
[10.1145/3716550.3722029](https://doi.org/10.1145/3716550.3722029).
- [10] **Andrew Chio**, Russell Bent, Kaarthik Sundar, and Nalini Venkatasubramanian. “Demo Abstract: SEQUIN: A Network Science and Physics-based Approach to Identify Sequential Nk Attacks in Electric Power Grids”. In: *Proceedings of the ACM/IEEE 16th International Conference on Cyber-Physical Systems (with CPS-IoT Week 2025) (IC-CPS'25)*. 2025, pp. 1–2.
[10.1145/3716550.3725165](https://doi.org/10.1145/3716550.3725165).

- [9] **Andrew Chio**, Russell Bent, Andrey Y Lokhov, Jian Peng, and Nalini Venkatasubramanian. “Physics-based Pollutant Source Identification in Stormwater Systems”. In: *Proceedings of the 22nd European Control Conference (ECC’24)*. IEEE. 2024, pp. 19–25.
[10.23919/ECC64448.2024.10591142.](#)
- [8] **Andrew Chio**, Jian Peng, and Nalini Venkatasubramanian. “STEP: toward a semantics-aware framework for monitoring community-scale infrastructure”. In: *Data-Centric Engineering (DCE) 5* (2024), e44.
JIF: 3.6 JCR2022.
[10.1017/dce.2024.32.](#)
- [7] Guoxi Wang, Ryan Hildebrant, **Andrew Chio**, Sharad Mehrotra, and Nalini Venkatasubramanian. “BatchIT: Intelligent and Efficient Batching for IoT Workloads at the Edge”. In: *IEEE Network Operations and Management Symposium (NOMS’24)*. IEEE. 2024, pp. 1–7.
B CORE2023.
[10.1109/NOMS59830.2024.10575298.](#)
- [6] **Andrew Chio**, Daokun Jiang, Peeyush Gupta, Georgios Bouloukakis, Roberto Yus, Sharad Mehrotra, and Nalini Venkatasubramanian. “SmartSPEC: A framework to generate customizable, semantics-based smart space datasets”. In: *Pervasive and Mobile Computing (PMC) 93* (2023), p. 101809.
JIF: 4.3 JCR2023 ,
[10.1016/j.pmcj.2023.101809.](#)
- [5] **Andrew Chio**, Jian Peng, and Nalini Venkatasubramanian. “STEP: Semantics-aware Sensor Placement for Monitoring Community-Scale Infrastructure”. In: *Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys’23)*. 2023, pp. 189–197.
A CORE2018.
[10.1145/3600100.3623752.](#)
- [4] **Andrew Chio**, Daokun Jiang, Peeyush Gupta, Georgios Bouloukakis, Roberto Yus, Sharad Mehrotra, and Nalini Venkatasubramanian. “SmartSPEC: Customizable Smart Space Datasets via Event-driven Simulations”. In: *Proceedings of the 20th IEEE International Conference on Pervasive Computing and Communications (PerCom’22)*. IEEE. 2022, pp. 152–162.
Best Paper Award, A CORE2021.*
[10.1109/PerCom53586.2022.9762405.](#)
- [3] **Andrew Chio**, Daokun Jiang, Peeyush Gupta, Georgios Bouloukakis, Roberto Yus, Sharad Mehrotra, and Nalini Venkatasubramanian. “Artifact: SmartSPEC: customizable smart space datasets via event-driven simulations”. In: *IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops’22)*. IEEE. 2022, pp. 183–184.
[10.1109/PerComWorkshops53856.2022.9767264.](#)
- [2] Yiming Lin, Daokun Jiang, Roberto Yus, Georgios Bouloukakis, **Andrew Chio**, Sharad Mehrotra, and Nalini Venkatasubramanian. “LOCATER: Cleaning WiFi Connectivity Datasets for Semantic Localization”. In: *Proceedings of the VLDB Endowment 14.3* (2020), pp. 329–341.
JIF: 3.8 JCR2024 ,
[10.5555/3430915.3442432.](#)
- [1] **Andrew Chio**, Georgios Bouloukakis, Cheng-Hsin Hsu, Sharad Mehrotra, and Nalini Venkatasubramanian. “Adaptive Mediation for Data Exchange in IoT Systems”. In: *Proceedings of the 18th Workshop on Adaptive and Reflexive Middleware (ARM’19)*. 2019, pp. 1–6.
[10.1145/3366612.3368122.](#)

Publications in Review/Preparation

- [3] **Andrew Chio**, Russell Bent, Kaarthik Sundar, and Nalini Venkatasubramanian. *Beyond Steady-State: A Decoupled Approach for N-k Interdiction with Immediate Automated System Response in Electric Power Grids*. In Preparation. 2026.
- [2] **Andrew Chio**, Jian Peng, and Nalini Venkatasubramanian. *SENTRY: Causality-based Anomaly Detection for Smart Cyber-Physical Infrastructures*. In Preparation. 2026.
- [1] **Andrew Chio** and Andrey Y. Lokhov. *Physics-Based Occupancy Inference from Building Temperature Data*. In Preparation. 2026.

Software Artifacts

- [4] **Andrew Chio**. *SEQUIN*. GitHub. 2025. [URL: https://github.com/andrewgchio/SEQUIN](https://github.com/andrewgchio/SEQUIN).
An analysis toolkit for exploring worst-case sequential and concurrent failures on power grids using an N-k interdiction model framework.
- [3] **Andrew Chio**. *Temporal N-k*. GitHub. 2025. [URL: https://github.com/andrewgchio/temporal-nk](https://github.com/andrewgchio/temporal-nk).
A toolkit for modeling and capturing cascading failures in power grids through an extended N-k interdiction model framework.
- [2] **Andrew Chio**. *STEP*. GitHub. 2023. [URL: https://github.com/andrewgchio/STEP](https://github.com/andrewgchio/STEP).
A dashboard for sensor placement, that leverages insights from structural, behavioral, and semantic aspects of a stormwater infrastructure for suitable deployments.
- [1] **Andrew Chio**. *SmartSPEC*. GitHub. 2022. [URL: https://github.com/andrewgchio/SmartSPEC](https://github.com/andrewgchio/SmartSPEC).
A smart space simulator and data generator that creates customizable smart space datasets using semantic models of spaces, people, events and sensors.

Other Talks and Presentations

- [5] **Andrew Chio**. *Enabling Real-Time Monitoring and Response for Community-Scale Infrastructure Networks*. CNLS Seminar Talk. 2026.
- [4] **Andrew Chio**, Russell Bent, Kaarthik Sundar, and Nalini Venkatasubramanian. *Augmenting N-k Interdiction Models with Temporal Aspects for Electric Grid Networks*. 2025 Grid Science Winter School and Conference. 2025.
- [3] **Andrew Chio**, Russell Bent, Kaarthik Sundar, and Nalini Venkatasubramanian. *SEQUIN: A Network Science and Physics-based Approach to Identify Sequential N-k Attacks in Electric Power Grids*. 2025 LANL Student Symposium. 2025.
- [2] **Andrew Chio**, Jian Peng, and Nalini Venkatasubramanian. *STEP: Semantics-Aware Sensor Placement for Monitoring Community-Scale Infrastructure*. ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, Posters and Demo Session. 2023.
- [1] **Andrew Chio**, Georgios Bouloukakakis, and Nalini Venkatasubramanian. *Distributed Mediator Placement for IoT Data Exchange Interoperability*. Surf-IoT Undergraduate Research Symposium. 2018.

News and Media

- [2] Johnny Loc Nguyen. *ALOHA: Inside a UCI Mathematics Professor's Effort to Maximize Hybrid Learning*. <https://www.compass.uci.edu/aloha-story/>. Apr. 2024.
- [1] Karen Phan. *Ph.D. Student Andrew Chio Named ARCS Scholar, UC National Lab In-Residence Fellow*. <https://ics.uci.edu/2022/12/06/ph-d-student-andrew-chio-named-arcs-scholar-uc-national-lab-in-residence-fellow/>. Dec. 2022.

Research Projects and Experiences

Principal Investigator

◇ LFRP In-Residence National Laboratory Graduate Fellows

University of California, Office of the President

Budget: \$139,536, (2022 – 2024)

PIs: Andrew Chio (*Lead PI*), Nalini Venkatasubramanian, Russell Bent

Project Title: *Integrating Model and Data-Driven Methods in IoT-enabled Resilient Infrastructure*

- This project focuses on combining physics-based models with data-driven techniques powered by IoT sensing and AI/ML to better understand, monitor, and enhance the resilience of critical infrastructure systems like water and urban networks. It aims to develop tools that use sparse, real-time sensor data to detect anomalies, infer their sources, and support informed decision-making for planning and operating more robust infrastructure.
- My Role: As the PI, I developed extended N-k interdiction models to identify worst-case sequential and cascading power grid failures that yield up to 47% more disruption. I also built physics-based inference models to accurately recover stormwater pollutant sources with near perfect inference (MSE ~ 0.02) and fast execution time (avg. ≤ 10 m)

Graduate Student Researcher / Participant

◇ Artificial Intelligence for Mission (ArtIMis) Project

Lab-Directed Research and Development, Los Alamos National Laboratory

- This project explores advances in applying artificial intelligence technologies to support core scientific and national security missions. It integrates AI and machine-learning methods into scientific workflows, computational tools, and mission-relevant applications to accelerate discovery, improve complex simulations and decision-making, and enhance capabilities across diverse research domains.
- My Role: Developing complex agents for the Universal Research and Scientific Assistant (URSA) project at LANL. Exploring and implementing methods for enhancing trust and consistency agents, such as adaptive self-consistency and verification.

◇ Enabling Human-Centered Digital Twins for Community Resilience

Collaborative Research, National Science Foundation

PIs: Nalini Venkatasubramanian, Sharad Mehrotra, Ronald Eguchi

- This joint US-Japan project focuses on enhancing community disaster resilience, particularly for vulnerable populations such as older adults, by developing a human-centered cyber-physical digital twin framework that integrates models of hazards, people, events, and processes. The framework aims to support interactive planning and scenario analysis, identify impacted populations in diverse hazard conditions, and embed real-time observational systems to improve preparedness and response decision-making.
- My Role: Developing new models to capture the geo-correlated failures in the power grid due to earthquake disasters, and the impact on nearby community structures (e.g., hospitals)

◇ SWADE: Smart Water Data Exchange

Smart and Connected Communities, National Science Foundation

PIs: Nalini Venkatasubramanian, Sharad Mehrotra, Shangping Ren, David Feldman, Ronald Eguchi

- The SWADE project is an interdisciplinary initiative to develop a unified Smart Water data-exchange framework that connects information from siloed water systems (drinking water, wastewater, and stormwater systems). It aims to improve how complex water infrastructure data is shared, analyzed, and used to enhance efficiency and resilience across the entire water cycle by engaging with community partners and stakeholders.
- My Role: Built a sensor placement decision-support tool that recommends practical deployments, improving system coverage and traceability by ~ 24 – 25% . I am also exploring adaptive, causality-based methods for pollutant discharge anomaly detection and control.

- ◇ **Privacy Cognizant IoT Environment for the Brandeis Program, DARPA**
PIs: Sharad Mehrotra, Alfred Kobsa, Nalini Venkatasubramanian
 - The project created an IoT smart space testbed, entitled TIPPERS, with a plug-n-play architecture to infuse diverse privacy enhancing technologies. Research explored novel ways to scaling privacy technologies to large data volume and usage scenario complexities, and to issues that arise in deploying such technologies in the real-world including challenges due to trust. TIPPERS has been transitioned to navy where it was a recipient of the NAVWAR innovations award in 2021 and part of the TRIDENT Warrior Exercises in 2019 and 2020.
 - My Role: Developed realistic simulations of human trajectories in naval ship environments for mission-critical and daily operations, achieving $1.4\times-4.4\times$ higher realism than baseline methods in Trident Warrior exercises (2019–2020). Also proposed a mediation-based architecture to support heterogeneous communications and protocols in IoT systems.

Teaching and Mentoring Experience

- ◇ **Undergraduate Research Mentor** Jun 2020 – Present
 - Aadi Agarwal, *UC Irvine*, Indiv. Study 2026
 - Andrew Yu, *UC Irvine*, Indiv. Study 2026
 - Max Peng, *UC Santa Barbara*, Indiv. Study 2025 – Present
 - Rohit De, *UC Irvine*, Indiv. Study 2025 - Present
 - Nhat Matthew Phan, *UC Irvine* → *M.S. at USC*, TIPPERS 2025
 - Badrish Ananth, *UC Irvine*, TIPPERS 2025
 - Miguel A Melo Ochoa, *San Diego State University*, IoT-SITY 2024
 - Anton Dimitriev, *Washington University in St. Louis* → *Ph.D. at UIUC*, IoT-SITY 2023
 - Christina Youn, *University of Notre Dame*, IoT-SITY 2020
- ◇ **Next Gen Pathways Mentor** Sep 2023 – Jun 2024
 - Aditya Karad (MS), *UC Irvine* → *Oracle, Databases*, 2023
 - Gaurav Ghati (MCS), *UC Irvine* → *Amazon Web Services, Redshift*, 2023
 - Sushmitha Jagannath (MCS), *UC Irvine* → *Goldman Sachs*, 2023
 - Haowei Xiong (MCS), *UC Irvine* → *TikTok*, 2023
 - Boying Lei (M.Eng), *UC Irvine*, 2023
- ◇ **Graduate Teaching Assistant** Sep 2019 – Jun 2021
Department of Computer Science, University of California, Irvine
 - ICS 46: Data Structures Implementation and Analysis (*Fall '19*)
 - ICS 33: Intermediate Programming (*Winter '20, Fall '20, Winter '21, Spring '21*)
 - CS 143A: Principles of Operating Systems (*Spring '20*)
- ◇ **Undergraduate CLAP Learning Assistant** Sep 2017 – Jun 2019
Certified Learning Assistants Program, University of California, Irvine
 - Math 1B: Pre-Calculus II (*Fall '18, Winter '19*)
 - Math 2B: Single-Variable Calculus II (*Spring '19*)
 - Physics 7LC: Classical Physics Lab (*Fall '17, Winter '18, Spring '18, Fall '18, Winter '19, Spring '19*)
- ◇ **Undergraduate Reader** Sep 2017 – Dec 2018
Department of Computer Science, University of California, Irvine
 - ICS 46: Data Structure Implementation and Analysis (*Fall '17, Winter '18, Fall '17*)

Professional Service

◇ Technical Program Committee Member

- IEEE International Conference on Pervasive Computing and Communications (PerCom): 2023

◇ Artifact Evaluation Committee Member

- IEEE International Conference on Software Architecture (ICSA): 2024
- IEEE International Conference on Pervasive Computing and Communications (PerCom): 2026

◇ Panel Moderator

- ACM Middleware Doctoral Symposium: 2025

◇ Panelist

- UCI Diverse Educational Community and Doctoral Experience (DECADE): 2025
- UC-NL LFRP Graduate Fellows Panel: 2024, 2025

◇ External Reviewer

- ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS): 2021, 2025, 2026
- ACM Transactions on Cyber Physical Systems (TCPS): 2025, 2026
- ACM/IFIP International Middleware Conference (Middleware): 2020, 2021, 2024, 2025
- ACM International Conference on Distributed and Event-based Systems (DEBS): 2022, 2025
- IEEE International Conference on Smart Computing (SMARTCOMP): 2020, 2022, 2023
- Journal of Ambient Intelligence and Smart Environments (JAISE): 2023
- Sustainable Computing: Informatics and Systems: 2025
- IEEE Internet of Things Journal: 2026

◇ Student Volunteer

- IEEE International Conference on Smart Computing (SMARTCOMP): 2021
- IEEE Mobile Data Management (MDM): 2025

◇ Professional Memberships

- ACM Member
- IEEE Member